

CS F212 DATABASE SYSTEMS 2024-25

LAB 3 EXERCISE

(GROUP BY, HAVING AND SET OPERATIONS)

1 - In **alphabetically ascending order**, display all the last names shared by more than one employee.

```
+-----+
| lastName |
+-----+
| Bondur   |
| Firrelli |
| Patterson |
+-----+
```

2 - Assume that any employee to whom more than 5 people report is considered a 'supervisor'. In **ascending order**, display the ID of every such 'supervisor' and the number of employees reporting to them.

```
+-----+-----+
| supervisor | no_employees |
+-----+-----+
|          1102 |          6 |
|          1143 |          6 |
+-----+-----+
```

3 - Among all countries with an average credit limit strictly less than 80000, display the one with the second-largest average credit limit. The field `country` is in the `customers` table.

```
+-----+-----+
| country | avg credit |
+-----+-----+
| France  | 77691.666667 |
+-----+-----+
```

4 - Display the two most profitable product lines (in the `products` table) in descending order of their total profit, which is given by:

```
+-----+-----+
| productLine | total_profit |
+-----+-----+
| Classic Cars | 11572218.67 |
| Vintage Cars | 4863250.86 |
+-----+-----+
```

5 - List the cities where both offices and customers reside. Make sure the order of display of your output matches with the below output.

```
+-----+
| city  |
+-----+
| San Francisco |
| Boston      |
| NYC         |
| Paris       |
| London      |
+-----+
```

6 - Find the employee number(s) to whom other employees report and also are assigned as a Sales Rep to a customer.

```
+-----+
| reportsTo |
+-----+
|      1621 |
+-----+
```

7 - Find the number of Sales Rep and VP Sales. Make sure the order of display of your output matches with the below output.

```
+-----+-----+
| jobTitle | C |
+-----+-----+
| Sales Rep | 17 |
| VP Sales  | 1  |
+-----+-----+
```

8 - Display the date when the maximum number of late orders was made. A late order is one that is shipped **on or after** the date on which it was required.

```
+-----+
| year |
+-----+
| 2004 |
+-----+
```

9 - Find the number of customers who did not order before May 1st, 2005.

```
+-----+
|Customers|
```

24

10 - Find the number of customers who have never ordered product number S72_3212.

Number
98